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# Qualitative Reasoning

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## 35.1 Introduction

Qualitative reasoning is the area of artificial intelligence (AI) that creates representations for continuous aspects of the world, such as space, time, and quantity, which support reasoning with very little information. Typically, it has focused on scientific and engineering domains, hence its other common name, qualitative physics. It is motivated by two observations. First, people draw useful and subtle conclusions about the physical world without equations. In our daily lives, we figure out what is happening around us and how we can affect it, working with far less data, and less precise data, than would be required to use traditional, purely quantitative methods. Creating software for robots that operate in unconstrained environments and modeling human cognition require understanding how this can be done. Second, scientists and engineers appear to use qualitative reasoning when initially understanding a problem, when setting up more formal methods to solve particular problems, and when interpreting the results of quantitative simulations, calculations, or measurements. Thus, advances in qualitative reasoning should lead to the creation of more flexible software that can help engineers and scientists solve problems.

Qualitative reasoning began with de Kleer's investigation on how qualitative and quantitative knowledge interacted in solving a subset of simple textbook mechanics problems (de Kleer, 1977). After roughly a decade of initial explorations, the potential for important industrial applications led to a surge

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of interest in the mid-1980s, and the area grew steadily, with rapid progress. There have been a variety of successful applications, including supervisory process control, design, failure modes and effects analysis (FMEA), and educational software.

Given its demonstrated utility in applications and its importance in understanding human cognition, work in qualitative modeling will continue to remain an important area in AI.

This chapter first surveys the state of the art in qualitative representations and in qualitative reasoning techniques. The application of these techniques to various problems is discussed subsequently.

## 35.2 Qualitative Representations

As is the case with many other knowledge representation issues, there is no single best qualitative representation. Instead, there exists a spectrum of choices, each with its own advantages and disadvantages for particular tasks. What all of them have in common is that they provide notations for describing and reasoning about continuous properties of the physical world. Two key issues in qualitative representation are *resolution* and *compositionality*. We discuss each in turn.

Resolution concerns the level of information detail in a representation. Resolution is an issue because one goal of qualitative reasoning is to understand how little information suffices to draw useful conclusions. Low-resolution information is available more often than precise information ("the car heading toward us is slowing down" vs. "the derivative of the car's velocity along the line connecting us is −28 km/h/s"), but conclusions drawn with low-resolution information are often ambiguous. The role of ambiguity is important: the prediction of alternative futures (i.e., "the car will hit us" vs. "the car won't hit us") suggests that we may need to gather more information, analyze the matter more deeply, or take action, depending on what alternatives our qualitative reasoning uncovers. High-resolution information is often needed to draw particular conclusions (i.e., a finite element analysis of heat flow within a tablet computer design to ensure that the CPU will not cook the battery), but qualitative reasoning with low-resolution representations reveals what the interesting questions are. Qualitative representations comprise one form of tacit knowledge that people, ranging from the person on the street to scientists and engineers, use to make sense of the world.

Compositionality concerns the ability to combine representations for different aspects of a phenomenon or system to create a representation of the phenomenon or system as a whole. Compositionality is an issue because one goal of qualitative reasoning is to formalize the modeling process itself. Many of today's AI systems are based on handcrafted knowledge bases that express information about a specific artifact or system needed to carry out a particular narrow range of tasks involving it. By contrast, a substantial component of the knowledge of scientists and engineers consists of principles and laws that are broadly applicable, both with respect to the number of systems they explain and the kinds of tasks they are relevant for. Qualitative reasoning has developed ideas and organizing techniques for knowledge bases with similar expressive and inferential power, called *domain theories*.

The remainder of this section surveys the fundamental representations used in qualitative reasoning for quantity, mathematical relationships, modeling assumptions, causality, space, and time.

### 35.2.1 Representing Quantity

Qualitative reasoning has explored trade-offs in representations for continuous parameters ranging in resolution from sign algebras to the hyperreals. Most of the research effort has gone into understanding the properties of low-resolution representations because the properties of high-resolution representations tend to already be well understood due to work in mathematics.

The lowest resolution representation for continuous parameters is the status abstraction, which represents a quantity by whether or not it is normal. It is a useful representation for certain diagnosis and monitoring tasks because it is the weakest representation that can express the difference between

something working and not working. The next step in resolution is the sign algebra, which represents continuous parameters as either −, +, or 0, according to whether the sign of the underlying continuous parameter is negative, positive, or zero. The sign algebra is surprisingly powerful: because a parameter's derivatives are themselves parameters whose values can be represented as signs, some of the main results of the differential calculus (e.g., the mean value theorem) can be applied to reasoning about sign values (de Kleer and Brown, 1984). This allows sign algebras to be used for qualitative reasoning about dynamics, including expressing properties such as oscillation and stability. The sign algebra is the weakest representation that supports such reasoning.

Representing continuous values via sets of ordinal relations (also known as the *quantity space* representation) is the next step up in resolution (Forbus, 1984). For example, the temperature of a fluid might be represented in terms of its relationship between the freezing point and boiling point of the material that comprises it. Like the sign algebra, quantity spaces are expressive enough to support qualitative reasoning about dynamics. (The sign algebra can be modeled by a quantity space with only a single point of comparison, zero.) Unlike the sign algebra, which draws values from a fixed finite algebraic structure, quantity spaces provide variable resolution because new points of comparison can be added to refine values. The temperature of water in a kettle on a stove, for instance, will likely be defined in terms of its relationship with the temperature of the stove as well as its freezing and boiling points. There are two kinds of comparison points used in defining quantity spaces. *Limit points* are derived from general properties of a domain as applicable to a specific situation. Continuing with the kettle example, the particular ordinal relationships used were chosen because they determine whether or not the physical processes of freezing, boiling, and heat flow occur in that situation. The precise numerical value of limit points can change over time (e.g., the boiling point of a fluid is a function of its pressure). *Landmark values* are constant points of comparison introduced during reasoning to provide additional resolution (Kuipers, 1994). To ascertain whether an oscillating system is overdamped, underdamped, or critically damped, for instance, requires comparing successive peak values. Noting the peak value of a particular cycle as a landmark value, and comparing it to the landmarks generated for successive cycles in the behavior, provides a way of making this inference.

Intervals are a well-known variable-resolution representation for numerical values and have been heavily used in qualitative reasoning. A quantity space can be thought of as partial information about a set of intervals. If we have complete information about the ordinal relationships between limit points and landmark values, these comparisons define a set of intervals that partition a parameter's value. This natural mapping between quantity spaces and intervals has been exploited by a variety of systems that use intervals whose end points are known numerical values to refine predictions produced by purely qualitative reasoning. Fuzzy intervals have also been used in similar ways, for example, in reasoning about control systems.

Order of magnitude representations stratify values according to some notion of scale. They can be important in resolving ambiguities and in simplifying models because they enable reasoning about what phenomena and effects can safely be ignored in a given situation. For instance, heat losses from turbines are generally ignored in the early stages of power plant design, because the energy lost is very small relative to the energy being produced. Several stratification techniques have been used in the literature, including hyperreal numbers, numerical thresholds, and logarithmic scales. Three issues faced by all these formalisms are (1) the conditions under which many small effects can combine to produce a significant effect, (2) the soundness of the reasoning supported by the formalism, and (3) the efficiency of using them (Travé-Massuyès et al., 2003).

Although many qualitative representations of number use the reals as their basis, another important basis for qualitative representations of number is finite algebras. One motivation for using finite algebras is that observations are often naturally categorized into a finite set of labels (i.e., very small, small, normal, large, very large). Research on such algebras is aimed at solving problems such as how to increase the compositionality of such representations (e.g., how to propagate information across different resolution scales).

### 35.2.2 Representing Mathematical Relationships

Like number, a variety of qualitative representations of mathematical relationships have been developed, often by adopting and adapting systems developed in mathematics. Abstractions of the analytic functions are commonly used to provide the lower resolution and compositionality desired. For example, confluences are differential equations over the sign algebra (de Kleer and Brown, 1984). An equation such as $V = I\,R$ can be expressed as the confluence

$$[V] = [I] + [R]$$

where $[Q]$ denotes taking the sign of $Q$. Differential equations can also be expressed in this manner, for instance,

$$[F] = \partial v$$

which is a qualitative version of $F = M\,A$ (assuming $M$ is always positive). Thus, any system of algebraic and differential equations with respect to time can be described as a set of confluences.

Many of the algebraic operations taken for granted in manipulating analytic functions over the reals are not valid in weak algebras (Struss, 1988). Because qualitative relationships are most often used to propagate information, this is not a serious limitation. In situations where algebraic solutions themselves are desirable, mixed representations that combine algebraic operations over the reals and move to qualitative abstractions when appropriate provide a useful approach (Williams, 1991).

Another low-resolution representation of equations uses monotonic functions over particular ranges; that is,

```
M+(acceleration, force)
```

states that acceleration depends only on the force, and the function relating them is increasing monotonic (Kuipers, 1994). Compositionality is achieved using *qualitative proportionalities* (Forbus, 1984) to express partial information about functional dependency; for example,

```
acceleration αQ+ force
```

states that acceleration depends on force and is increasing monotonic in its dependence on force but may depend on other factors as well. Additional constraints on the function that determines force can be added by additional qualitative proportionalities; for example,

```
acceleration αQ- mass
```

states that acceleration also depends on mass and is decreasing monotonic in this dependence. Qualitative proportionalities must be combined via closed-world assumptions to ascertain all the effects on a quantity. Similar primitives can be defined for expressing relationships involving derivatives, to define a complete language of compositional qualitative mathematics for ordinary differential equations. As with confluences, few algebraic operations are valid for combining monotonic functions. Composition of functions of identical sign is safe; that is,

$$\mathrm{M+}(f,g) \wedge \mathrm{M+}(g,h) \Rightarrow \mathrm{M+}(f,h)$$

In addition to resolution and compositionality, another issue arising in qualitative representations of mathematical relationships is causality. There are three common views on how mathematical relationships interact with the causal relationships people use in commonsense reasoning. One view is that

there is no relationship between them. The second view is that mathematical relationships should be expressed with primitives that also make causal implications. For example, qualitative proportionalities include a causal interpretation, that is, a change in force causes a change in acceleration, but not the other way around. The third view is that acausal mathematical relationships give rise to causal relationships via the particular process of using them. For example, confluences have no built-in causal direction but are used in causal reasoning by identifying the flow of information through them while reasoning with a presumed flow of causality in the physical system they model. One method for imposing causality on a set of acausal constraint equations is by computing a causal ordering (Iwasaki and Simon, 1994) that imposes directionality on a set of equations, starting from variables considered to be exogenous within the system.

Each view of causality has its merits. For tasks where causality is truly irrelevant, ignoring causality can be a reasonable approach. To create software that can span the range of human commonsense reasoning, something like a combination of the second and third views appears necessary because the appropriate notion of causality varies (Forbus and Gentner, 1986). In reasoning about chemical phenomena, for instance, changes in concentration are always caused by changes in the amounts of the constituent parts and never the other way around. In electronics, on the other hand, it is often convenient to consider voltage changes as being caused by changes in current in one part of a circuit and to consider current changes as being caused by changes in voltage in another part of the same circuit.

### 35.2.3 Ontology

Ontology concerns how to carve up the world, that is, what kinds of things there are and what sorts of relationships can hold between them. Ontology is central to qualitative modeling because one of its main goals is formalizing the art of building models of continuous systems. A key choice in any act of modeling is figuring out how to construe the situation or system to be modeled in terms of the available models for classes of entities and phenomena. No single ontology will suffice for the span of reasoning about continuous systems that people do. What has been developed instead is a catalog of ontologies, describing their properties and interrelationships and specifying conditions under which each is appropriate. While several ontologies are currently well understood, the catalog still contains many gaps.

An example of ontologies will make this point clearer. Consider the representation of liquids. Broadly speaking, the major distinction in reasoning about fluids is whether one individuates fluid according to a particular collection of particles or by location (Hayes, 1985). The former are called Eulerian, or piece of stuff, ontologies. The latter are called Lagrangian, or contained stuff, ontologies. The contained stuff view of liquids is used when one treats a river as a stable entity, although the particular set of molecules that comprises it is changing constantly. The piece of stuff view of liquids is used when one thinks about the changes in a fluid as it flows through a steady-state system, such as a working refrigerator. Ontologies multiply as we try to capture more aspects of human reasoning. For instance, the piece of stuff ontology can be further divided into three cases, each with its own rules of inference: (1) molecular collections, which describe the progress of an arbitrary piece of fluid that is small enough to never split apart but large enough to have extensive properties; (2) slices, which, like molecular collections, never subdivide but unlike them are large enough to interact directly with their surroundings; and (3) pieces of stuff large enough to be split into several pieces (e.g., an oil slick). Similarly, the contained stuff ontology can be further specialized according to whether or not individuation occurs simply by container (abstract-contained stuffs) or by a particular set of containing surfaces (bounded stuffs). Abstract-contained stuffs provide a low-resolution ontology appropriate for reasoning about system-level properties in complex systems (e.g., the changes over time in a lubricating oil subsystem in a propulsion plant), whereas bounded stuffs contain the geometric information needed to reason about the interactions of fluids and shape in systems such as pumps and internal combustion engines.

Cutting across the ontologies for particular physical domains are systems of organization for classes of ontologies. The most commonly used ontologies are the device ontology (de Kleer and Brown, 1984)

and the process ontology (Forbus, 1984). The device ontology is inspired by network theory and system dynamics. Like those formalisms, it construes continuous systems as networks of devices whose interactions occur solely through a fixed set of ports. Unlike those formalisms, it provides the ability to write and reason automatically with device models whose governing equations can change over time.

The process ontology is inspired by studies of human mental models and observations of practice in thermodynamics and chemical engineering. It construes continuous systems as consisting of entities whose changes are caused by processes. Process ontologies thus postulate a separate ontological category for causal mechanisms, unlike device ontologies, where causality arises solely from the interaction of the parts. Another difference between the two classes of ontologies is that in the device ontology the system of devices and connections is fixed over time, whereas in the process ontology entities and processes can come into existence and vanish over time. Each is appropriate in different contexts: for most purposes, an electronic circuit is best modeled as a network of devices, whereas a chemical plant is best modeled as a collection of interacting processes.

### 35.2.4 State, Time, and Behaviors

A qualitative state is a set of propositions that characterize a qualitatively distinct behavior of a system. A qualitative state describing a falling ball, for instance, would include information about what physical processes are occurring (e.g., motion downward, acceleration due to gravity) and how the parameters of the ball are changing (e.g., its position is getting lower and its downward velocity is getting larger). A qualitative state can abstractly represent an infinite number of quantitative states: although the position and velocity of the ball are different at each distinct moment during its fall, until the ball collides with the ground, the qualitative state of its motion is unchanged.

Qualitative representations can be used to partition behavior into natural units. For instance, the time over which the state of the ball falling holds is naturally thought of as an interval, ending when the ball collides with the ground. The collision itself can be described as yet another qualitative state, and the fact that falling leads to a collision with the ground can be represented via a transition between the two states. If the ball has a nonzero horizontal velocity and there is some obstacle in its direction of travel, another possible behavior is that the ball will collide with that object instead of the ground. In general, a qualitative state can have transitions to several next states, reflecting ambiguity in the qualitative representations. Returning to our ball example, and assuming that no collisions with obstacles occur, notice that the qualitative state of the ball falling occurs again once the ball has reached its maximum height after the collision. If continuous values are represented by quantity spaces and the sources of comparisons are limit points, then a finite set of qualitative states is sufficient to describe every possible behavior of a system. A collection of such qualitative states and transitions is called an *envisionment* (de Kleer, 1977). Many interesting dynamical conclusions can be drawn from an envisionment. For instance, oscillations correspond to cycles of states. Unfortunately, the fixed resolution provided by limit points is not sufficient for other dynamical conclusions, such as ascertaining whether or not the ball's oscillation is damped. If comparisons can include landmark values, such conclusions can sometimes be drawn, for example, by comparing the maximum height on one bounce to the maximum height obtained on the next bounce. The cost of introducing landmark values is that the envisionment no longer need be finite; every cycle in a corresponding fixed-resolution envisionment could give rise to an infinite number of qualitative states in an envisionment with landmarks.

A sequence of qualitative states occurring over a particular span of time is called a *behavior*. Behaviors can be described using purely qualitative knowledge, purely quantitative knowledge, or a mixture of both. If every continuous parameter is quantitative, the numerical aspects of behaviors coincide with the notion of trajectory in a state-space model. If qualitative representations of parameters are used, a single behavior can represent a family of trajectories through state space.

An idea closely related to behaviors is *histories* (Hayes, 1985). Histories can be viewed as local behaviors, that is, how a single individual or property varies through time. A behavior is equivalent to a

global history, that is, the union of all the histories for the participating individuals. The distinction is important for two reasons. First, histories are the dual of situations in the situation calculus; histories are bounded spatially and extended temporally, whereas situations are bounded temporally and extended spatially. Using histories avoids the frame problem, instead trading it for the more tractable problems of generating histories locally and determining how they interact when they intersect in space and time. The second reason is that history-based simulation algorithms can be more efficient than state-based simulation algorithms because no commitments need to be made concerning irrelevant information.

In a correct envisionment, every possible behavior of the physical system corresponds to some path through the envisionment. Because envisionments reflect only local constraints, the converse is not true; that is, an arbitrary path through an envisionment may not represent a physically possible behavior. When using an envisionment to generate hypotheses about possible behaviors, this can lead to over-generation. Paths must be tested against global constraints, such as energy conservation, to ensure their physical validity, or filtered by some other means. When using an envisionment to explain observed behaviors, this limitation is less serious, since the observed behavior, by its very nature, must be physically possible. A more serious limitation is that envisionments are often exponential in the size of the system being modeled. This means that, in practice, envisionments often are not generated explicitly and, instead, possible behaviors are searched in ways similar to those used in other areas of AI.

Many tasks require integrating qualitative states with other models of time, such as numerical models. Including precise information (e.g., algebraic expressions or floating-point numbers) about the end points of intervals in a history does not change their essential character.

## 35.3 Space and Shape

Qualitative representations of shape and space play an important role in spatial cognition because they provide a bridge between the perceptual and the conceptual. By discretizing continuous space, they make it amenable to symbolic reasoning. As with qualitative representations of 1D parameters, task constraints govern the choice of qualitative representation. However, problem-independent purely qualitative spatial representations suffice for fewer tasks than in the 1D case, because of the increased ambiguity in higher dimensions (Forbus et al., 1991). Consider, for example, deciding whether a protrusion can fit snugly inside a hole. If we have detailed information about their shapes, we can derive an answer. If we consider a particular set of protrusions and a particular set of holes, we can construct a qualitative representation of these particular protrusions and holes that would allow us to derive whether or not a specific pair would fit, based on their relative sizes. But if we first compute a qualitative representation for each protrusion and hole in isolation, in general the rules of inference that can be derived for this problem will be very weak. Work in qualitative spatial representations thus tends to take two approaches. The first approach is to explore what aspects do lend themselves to purely qualitative representations. The second approach is to use a quantitative representation as a starting point and compute problem-specific qualitative representations to reason with. We summarize each in turn.

An impressive catalog of qualitative spatial representations have been developed that can be used to reason about particular aspects of shape and space (Cohn and Renz, 2008). The general approach is to identify a set of relationships that provide natural decompositions of a spatial property, such as connectivity or orientation, and define calculi over these relationships to support inference. For example, the eight relationship versions of the region connection calculus (RCC) include disjoint, edge connected, partially overlapping, tangential proper part (and its inverse), non-tangential proper part (and its inverse), and equal as relationships. A transitivity table supports inference, that is, if a bird is inside a cage and a cat is outside the cage, then the bird and the cat are disjoint (and unless the connectivity changes, the bird is safe). Note that the sequence of relationships earlier also captures

some aspects of the relationships between two objects that are moving: they are apart, they touch, one enters the other, etc. Such physical interpretations have enabled such calculi to be used in reasoning about change and in interpreting visual information (Sridhar et al., 2010). The calculi vary in terms of expressiveness and efficiency: some have been shown to be NP-complete, but often tractable subsets can be found.

The use of quantitative representations to ground qualitative spatial reasoning can be viewed as a model of the ways humans use diagrams and models in spatial reasoning. For this reason such work is also known as *diagrammatic reasoning* (Glasgow et al., 1995). One form of diagram representation is the occupancy array that encodes the location of an object by cells in a (2D or 3D) grid. These representations simplify the calculation of spatial relationships between objects (e.g., whether or not one object is above another), albeit at the cost of making the object's shape implicit. Another form of diagram representation uses symbolic structures with quantitative properties, for example, numerical, algebraic, or interval (e.g., Forbus et al., 1991). These representations simplify calculations involving shape and spatial relationships, without the scaling and resolution problems that sometimes arise in array representations. However, they require a set of primitive shape elements that spans all the possible shapes of interest, and identifying such sets for particular tasks can be difficult. For instance, many intuitively natural sets of shape primitives are not closed with respect to their complement, which can make characterizing free space difficult.

Diagram representations are used for qualitative spatial reasoning in two ways. The first is as a decision procedure for spatial questions. This mimics one of the roles diagrams play in human perception. Often, these operations are combined with domain-specific reasoning procedures to produce an analog style of inference, where, for instance, the effects of perturbations on a structure are mapped into the diagram, the effect on the shapes in the diagram noted, and the results mapped back into a physical interpretation. The second way uses the diagram to construct a problem-specific qualitative vocabulary, imposing new spatial entities representing physical properties, such as the maximum height a ball can reach or regions of free space that can contain a motion. This is the *metric diagram/place vocabulary* model of qualitative spatial reasoning.

Representing and reasoning about kinematic mechanisms was one of the early successes in qualitative spatial reasoning. The possible motions of objects are represented by qualitative regions in the configuration space representing the legitimate positions of parts of mechanisms (Faltings, 1990). Whereas, in principle, a single high-dimensional configuration space could be used to represent a mechanism's possible motions (each dimension corresponding to a degree of freedom of a part of the mechanism), in practice a collection of configuration spaces, one 2D space for each pair of parts that can interact, is used. These techniques suffice to analyze a wide variety of kinematic mechanisms and can support design (Sacks and Joscowicz, 2010).

Another important class of spatial representations concerns qualitative representations of spatially distributed phenomena, such as flow structures and regions in phase space. These models use techniques from computer vision to recognize or impose qualitative structure on a continuous field of information, gleaned from numerical simulation or scientific data. This qualitative structure, combined with domain-specific models of how such structures tie to the underlying physics, enables them to interpret physical phenomena in much the same way that a scientist examining the data would (e.g., Yip, 1991; Huang and Zhao, 2000).

An important recent trend is using rich, real-world data as input for qualitative spatial reasoning. For example, several systems provide some of the naturalness of sketching by performing qualitative reasoning on spatial data input as digital ink, for tasks like mechanical design (Stahovich et al., 1998), reasoning about student's sketches in geoscience and engineering design education (Forbus et al., 2011), and communicating tactics to robot soccer teams (Gspandl et al., 2010). Qualitative representations are being used in computer vision as well, for example, as a means of combining dynamic scenes across time to interpret events (e.g., Sridhar et al., 2010).

### 35.3.1 Compositional Modeling, Domain Theories, and Modeling Assumptions

There is almost never a single correct model for a complex physical system. Most systems can be modeled in a variety of ways, and different tasks can require different types of models. The creation of a system model for a specific purpose is still something of an art. Qualitative reasoning has developed formalisms that combine logic and mathematics with qualitative representations to help automate the process of creating and refining models. The compositional modeling methodology (Falkenhainer and Forbus, 1991), which has become standard in qualitative reasoning, works like this: models are created from domain theories, which describe the kinds of entities and phenomena that can occur in a physical domain. A domain theory consists of a set of *model fragments*, each describing a particular aspect of the domain. Creating a model is accomplished by instantiating an appropriate subset of model fragments, given some initial specification of the system (e.g., the propositional equivalent of a blueprint) and information about the task to be performed. Reasoning about appropriateness involves the use of modeling assumptions. Modeling assumptions are the control knowledge used to reason about the validity or appropriateness of using model fragments. Modeling assumptions are used to express the relevance of model fragments. Logical constraints between modeling assumptions comprise an important component of a domain theory.

An example of a modeling assumption is assuming that a turbine is isentropic. Here is a model fragment that illustrates how this assumption is used:

```
(defEquation Isentropic-Turbine
    ((turbine ?g ?in ?out)(isentropic ?g))
    (= (spec-s ?in) (spec-s ?out)))
```

In other words, when a turbine is isentropic, the specific entropy of its inlet and outlet are equal. Other knowledge in the domain theory puts constraints on the predicate isentropic,

```
(for-all (?self (turbine ?self))
   (iff (= (nu-isentropic ?self) 1.0) (isentropic ?self)))
```

That is, a turbine is isentropic exactly when its isentropic thermal efficiency is 1. Although no real turbine is isentropic, assuming that turbines are isentropic simplifies early analyses when creating a new design. In later design phases, when tighter performance bounds are required, this assumption is retracted and the impact of particular values for the turbine's isentropic thermal efficiency is explored. The consequences of choosing particular modeling assumptions can be quite complex; the fragments shown here are less than one-fourth the knowledge expressing the consequences of assuming that a turbine is isentropic in a typical knowledge base.

Modeling assumptions can be classified in a variety of ways. An *ontological assumption* describes which ontology should be used in an analysis. For instance, reasoning about the pressure at the bottom of a swimming pool is most simply performed using a contained stuff representation, whereas describing the location of an oil spill is most easily performed using a piece of stuff representation. A *perspective assumption* describes which subset of phenomena operating in a system will be the subject. For example, in analyzing a steam plant, one might focus on a fluid perspective, a thermal perspective, or both at once. A *granularity assumption* describes how much detail is included in an analysis. Ignoring the implementation details of subsystems, for instance, is useful in the conceptual design of an artifact, but the same implementation details may be critical for troubleshooting that artifact. The relationships between these classes of assumptions can be complicated and domain dependent; for instance, it makes no sense to include a model of a heating coil (a choice of granularity) if the analysis does not include thermal properties (a choice of perspective).

Relationships between modeling assumptions provide global structure to domain theories. Assumptions about the nature of this global structure can significantly impact the efficiency of model formulation, as discussed in the following. In principle, any logical constraint could be imposed between modeling assumptions. In practice, two kinds of constraints are the most common. The first are implications, such as one modeling assumption requiring or forbidding another. For example,

```
(for-all (?s (system ?s))
  (implies (consider (black-box ?s))
           (for-all (?p (part-of ?p ?s)) (not (consider ?p)))))
```

says that if one is considering a subsystem as a black box, then all of its parts should be ignored. Similarly,

```
(for-all (?l (contained-stuff ?l))
   (implies (consider (pressure ?l))
            (consider (fluid-properties ?l))))
```

states that if an analysis requires considering something's pressure, then its fluid properties are relevant. The second kind of constraint between modeling assumptions is *assumption classes.* An assumption class expresses a choice required to create a coherent model under particular conditions. For example,

```
(defAssumptionClass (turbine ?self)
  (isentropic ?self) (not (isentropic ?self)))
```

states that when something is modeled as a turbine, any coherent model including it must make a choice about whether or not it is modeled as isentropic. The choice may be constrained by the data so far (e.g., different entrance and exit specific entropies), or it may be an assumption that must be made in order to complete the model. The set of choices need not be binary. For each valid assumption class, exactly one of the choices it presents must be included in the model.

## 35.4 Qualitative Reasoning Techniques

A variety of qualitative reasoning techniques have been developed that use the qualitative representations just outlined.

### 35.4.1 Model Formulation

Methods for automatically creating models for a specific task are one of the hallmark contributions of qualitative reasoning. These methods formalize knowledge and skills typically left implicit by most of traditional mathematics and engineering.

The simplest model formulation algorithm is to instantiate every possible model fragment from a domain theory, given a propositional representation of the particular scenario to be reasoned about. This algorithm is adequate when the domain theory is very focused and thus does not contain much irrelevant information. It is inadequate for broad domain theories and fails completely for domain theories that include alternative and mutually incompatible perspectives (e.g., viewing a contained liquid as a finite object vs. an infinite source of liquid). It also fails to take task constraints into account. For example, it is possible, in principle, to analyze the cooling of a cup of coffee using quantum mechanics. Even if it were possible in practice to do so, for most tasks simpler models suffice. Just how simple a model can be and remain adequate depends on the task. If I want to know if the cup of coffee will still be drinkable after an hour, a qualitative model suffices to infer that its final temperature will be that of its surroundings. If I want to know its temperature within 5% after 12 min have passed, a macroscopic quantitative model is a better choice. In other words, the goal of model formulation is to create the simplest adequate model of a system for a given task.

More sophisticated model formulation algorithms search the space of modeling assumptions, because they control which aspects of the domain theory will be instantiated. The model formulation algorithm of Falkenhainer and Forbus (1991) instantiated all potentially relevant model fragments and used an assumption-based truth maintenance system to find all legal combinations of modeling assumptions that sufficed to form a model that could answer a given query. The simplicity criterion used was to minimize the number of modeling assumptions. This algorithm is very simple and general but has two major drawbacks: (1) full instantiation can be very expensive, especially if only a small subset of the model fragments is eventually used; and (2) the number of consistent combinations of model fragments tends to be exponential for most problems. The rest of this section describes algorithms that overcome these problems.

Efficiency in model formulation can be gained by imposing additional structure on domain theories. Under at least one set of constraints, model formulation can be carried out in polynomial time (Nayak, 1994). The constraints are that (1) the domain theory can be divided into independent assumption classes and, (2) within each assumption class, the models can be organized by a (perhaps partial) simplicity ordering of a specific nature, forming a lattice of causal approximations. Nayak's algorithm computes a simplest model, in the sense of simplest within each local assumption class, but does not necessarily produce the globally simplest model.

Conditions that ensure the creation of coherent models, that is, models that include sufficient information to produce an answer of the desired form, provide powerful constraints on model formulation. For example, in generating "what-if" explanations of how a change in one parameter might affect particular other properties of the system, a model must include a complete causal chain connecting the changed parameter to the other parameters of interest. This insight can be used to treat model formulation as a best-first search for a set of model fragments providing the simplest complete causal chain (Rickel and Porter, 1994). A novel feature of this algorithm is that it also selects models at an appropriate timescale. It does this by choosing the slowest timescale phenomenon that provides a complete causal model, because this provides accurate answers that minimize extraneous detail.

As with other AI problems, knowledge can reduce search. One kind of knowledge that experienced modelers accumulate concerns the range of applicability of various modeling assumptions and strategies for how to reformulate when a given model proves inappropriate. Dynamically constructing qualitative representations from more detailed models, driven by task requirements, has been explored (Sachenbacher and Struss, 2005). Model formulation often is an iterative process. For instance, an initial qualitative model often is generated to identify the relevant phenomena, followed by the creation of a narrowly focused quantitative model to answer the questions at hand. Similarly, domain-specific error criterion can determine that a particular model's results are internally inconsistent, causing the reasoner to restart the search for a good model. One approach is to formalize model formulation as a dynamic preference constraint satisfaction problem, where more fine-grained criteria for model preference than "simplest" can be formalized and exploited (Keppens and Shen, 2004). Another promising approach is analogical model formulation, since similar models are likely to work in similar circumstances (Klenk et al., 2011).

### 35.4.2 Causal Reasoning

Causal reasoning explains an aspect of a situation in terms of others in such a way that the aspect being explained can be changed if so desired. For instance, a flat tire is caused by the air inside flowing out, either through the stem or through a leak. To refill the tire, we must both ensure that the stem provides a seal and that there are no leaks. Causal reasoning is thus at the heart of diagnostic reasoning as well as explanation generation.

The techniques used for causal reasoning depend on the particular notion of causality used, but they all share a common structure. First, causality involving factors within a state are identified. Second, how the properties of a state contribute to a transition (or transitions) to another state are identified, to extend the causal account over time. Because causal reasoning often involves qualitative simulation, we turn to simulation next.

### 35.4.3 Simulation

The new representations of quantity and mathematical relationships of qualitative reasoning expand the space of simulation techniques considerably. We start by considering varieties of purely qualitative simulation and then describe several simulation techniques that integrate qualitative and quantitative information.

Understanding *limit analysis*, the process of finding state transitions, is key to understanding qualitative simulation. Recall that a qualitative state consists of a set of propositions, some of them describing the values of continuous properties in the system. (For simplicity in this discussion, we will assume that these values are described as ordinal relations, although the same method works for sign representations and representations richer than ordinals.) Two observations are critical: (1) the phenomena that cause changes in a situation often depend on ordinal relationships between parameters of the situation, and (2) knowing just the sign of the derivatives of the parameters involved in these ordinal relationships suffices to predict how they might change over time. The effects of these changes, when calculated consistently, describe the possible transitions to other states.

An example will make this clearer. Consider again a pot of water sitting on a stove. Once the stove is turned on, heat begins to flow to the water in the pot because the stove's temperature is higher than that of the water. The causal relationship between the temperature inequality and the flow of heat means that to predict changes in the situation, we should figure out their derivatives and any other relevant ordinal relationships that might change as a result. In this qualitative state, the derivative of the water's temperature is positive, and the derivative of the stove's temperature is zero. Thus, one possible state change is that the water will reach thermal equilibrium with the stove and the flow of heat will stop. That is not the only possibility, of course. We know that boiling can occur if the temperature of the water begins to rise above its boiling temperature. That, too, is a possible transition that would end the state. Which of these transitions occurs depends on the relationship between the temperature of the stove and the boiling temperature of water.

This example illustrates several important features of limit analysis. First, surprisingly weak information (i.e., ordinal relations) suffices to draw important conclusions about broad patterns of physical behavior. Second, limit analysis with purely qualitative information is fundamentally ambiguous: it can identify what transitions might occur, but cannot by itself determine in all cases which transition will occur. Third, like other qualitative ambiguities, higher-resolution information can be brought in to resolve the ambiguities as needed. Returning to our example, any information sufficient to determine the ordinal relationship between the stove temperature and boiling suffices to resolve this ambiguity. If we are designing an electric kettle, for instance, we would use this ambiguity as a signal that we must ensure that the heating element's temperature is well above the boiling point; and if we are designing a drink warmer, its heating element should operate well below the boiling point.

Qualitative simulation algorithms vary along four dimensions: (1) their initial states, (2) what conditions they use to filter states or transitions, (3) whether or not they generate new landmarks, and (4) how much of the space of possible behaviors they explore. Envisioning is the process of generating an envisionment, that is, generating all possible behaviors. Two kinds of envisioning algorithms have been used in practice: attainable envisioners produce all states reachable from a set of initial states, and total envisioners produce a complete envisionment. Behavior generation algorithms start with a single initial state, generate landmark values, and use a variety of task-dependent constraints as filters and termination criteria (e.g., resource bounds, energy constraints).

Higher-resolution information can be integrated with qualitative simulation in several ways. One method for resolving ambiguities in behavior generation is to provide numerical envelopes to bound mathematical relationships. These envelopes can be dynamically refined to provide tighter situation-specific bounds. Such systems are called *semiquantitative* simulators (Kuipers, 1994).

A different approach to integration is to use qualitative reasoning to automatically construct a numerical simulator that has integrated explanation facilities. These *self-explanatory simulators* (Forbus and

Falkenhainer, 1990) use traditional numerical simulation techniques to generate behaviors, which are also tracked qualitatively. The concurrently evolving qualitative description of the behavior is used both in generating explanations and in ensuring that appropriate mathematical models are used when applicability thresholds are crossed. Self-explanatory simulators can be compiled in polynomial time for efficient execution, even on small computers, or created in an interpreted environment. They have been deployed mostly in education, both as a web-based medium and within classroom activities for middle school students.

### 35.4.4 Comparative Analysis

Comparative analysis answers a specific kind of "what-if" questions, namely, the changes that result from changing the value of a parameter in a situation. Given higher-resolution information, traditional analytic or numerical sensitivity analysis methods can be used to answer these questions; however (1) such reasoning is commonly carried out by people who have neither the data nor the expertise to carry out such analyses, and (2) purely quantitative techniques tend not to provide good explanations. Sometimes, purely qualitative information suffices to carry out such reasoning, using techniques such as exaggeration (Weld, 1990). Consider, for instance, the effect of increasing the mass of a block in a spring-block oscillator. If the mass were infinite, the block would not move at all, corresponding to an infinite period. Thus, we can conclude that increasing the mass of the block will increase the period of the oscillator.

### 35.4.5 Teleological Reasoning

Teleological reasoning connects the structure and behavior of a system to its goals. (By its goals, we are projecting the intent of its designer or the observer, because purposes often are ascribed to components of evolved systems.) To describe how something works entails ascribing a function to each of its parts and to explain how these functions together achieve the goals. Teleological reasoning is accomplished by a combination of abduction and recognition. Abduction is necessary because most components and behaviors can play several functional roles (de Kleer, 1984). A turbine, for instance, can be used to generate work in a power generation system and to expand a gas in a liquefaction system. Recognition is important because it explains patterns of function in a system in terms of known, commonly used abstractions. A complex power generation system with multiple stages of turbines and reheating and regeneration, for instance, still can be viewed as a Rankine cycle after the appropriate aggregation of physical processes involved in its operation (Everett, 1999).

### 35.4.6 Data Interpretation

There are two ways that the representations of qualitative reasoning have been used in data interpretation problems. The first is to explain a temporal sequence of measurements in terms of a sequence of qualitative states; the second is to create a qualitative model of spatially distributed numerical data, for example, phase spaces, by interpreting the results of empirical data measurements or successive numerical simulation experiments in terms of regions whose qualitative properties are constant. The underlying commonality in these problems is the use of qualitative descriptions of physical constraints to formulate compatibility constraints that prune the set of possible interpretations. We summarize each in turn.

In measurement interpretation tasks, numerical and symbolic data are partitioned into intervals, each of which can be explained by a qualitative state or sequence of qualitative states. Using precomputed envisionments or performing limit analysis online, possible transitions between states used as interpretations can be found for filtering purposes. Specifically, if a state $S_1$ is a possible interpretation for interval $I_1$, then at least one transition from $S_1$ must lead to a state that is an interpretation for the next interval.

This compatibility constraint, applied in both directions, can provide substantial pruning. Additional constraints that can be applied include the likelihood of particular states occurring, the likelihood of particular transitions occurring, and estimates of durations for particular states. Algorithms have been developed that can use all these constraints to maintain a single best interpretation of a set of incoming measurements that operate in polynomial time (de Coste, 1991).

In spatial data interpretation tasks, a physical experiment (e.g., Huang and Zhao, 2000) or numerical simulation (e.g., Yip, 1991) is used to gather information about the possible behaviors of a system given a set of initial parameters. The geometric patterns these behaviors form are described using vision techniques to create a qualitative characterization of the behavior. For example, initially, simulations are performed on a coarse grid to create an initial description of the system's phase space. This initial description is then used to guide additional numerical simulation experiments, using rules that express physical properties visually.

### 35.4.7 Spatial Reasoning

Reasoning with purely qualitative representations uses constraint satisfaction techniques to determine possible solutions to networks of relationships. The constraints are generally expressed as transitivity tables. When metric diagrams are used, processing techniques adapted from vision and robotics research are used to extract qualitative descriptions (Bailey-Kellogg and Zhao, 2003). Some reasoning proceeds purely within these new qualitative representations, while other tasks require the coordination of qualitative and diagrammatic representations. There is now a robust flow in the other direction, that is, vision and robotics researchers adopting qualitative representations because they are more robust to compute from the data and are more appropriate for many tasks (e.g., Kuipers and Byun, 1991; Sridhar et al., 2010). For example, qualitative spatial models computed from sensors are being used by teams in RoboCup Soccer (e.g., Schiffer et al., 2006).

## 35.5 Applications of Qualitative Reasoning

Qualitative reasoning began as a research enterprise in the 1980s, with successful fielded applications starting to appear by the early 1990s. For example, applications in supervisory process control (LeClair et al., 1989) have been successful enough to be embedded in commercial systems. Qualitative reasoning techniques have been used in the design of complex electromechanical systems, such as photocopiers, including the automatic generation of control software. A commercial tool using qualitative reasoning for FMEA in automobile electrical circuits was adopted by a major automobile manufacturer (Price, 2000). Thus, some applications of qualitative reasoning have been in routine use for over a decade, and more applications continue to be developed, with efforts expanding to include education and scientific modeling as well as engineering. Here we briefly summarize some of these research efforts.

### 35.5.1 Monitoring, Control, and Diagnosis

Monitoring, control, and diagnosis, although often treated as distinct problems, are in many applications deeply intertwined. Because these tasks also have deep theoretical commonalities, they are described together here. Monitoring a system requires summarizing its behavior at a level of description that is useful for taking action. Qualitative representations correspond to descriptions naturally applied by system operators and designers and thus can help provide new opportunities for automation. An important benefit of using qualitative representations is that the concepts the software uses can be made very similar to those of people who interact with software, thus potentially improving human-computer interfaces. Qualitative representations are important for control because qualitative distinctions provide criteria that make different control actions appropriate. Diagnosis tasks impose similar requirements. It is rarely beneficial to spend the resources required to construct a very detailed quantitative model of the

way a particular part has failed when the goal is to isolate a problem. Qualitative models often provide sufficient resolution for fault isolation. Qualitative models also provide the framework for organizing fault detection (i.e., noticing that a problem has occurred) and for working around a problem, even when these tasks require quantitative information (e.g., Flad et al., 2010).

An early commercial success of qualitative reasoning was in supervisory process control, where qualitative representations are used to provide more robust control than statistical process control in curing composite parts (LeClair et al., 1989). In the early stage of curing a composite part, the temperature of the furnace needs to be kept relatively low because the part is outgassing. Keeping the furnace low during the entire curing process is inefficient, however, because lower temperatures mean longer cure times. Therefore, it is more productive to keep temperature low until outgassing stops and then increase it to finish the cure process more quickly. Statistical process control methods use a combination of analytic models and empirical tests to figure out an optimal pattern of high/low cooking times. By incorporating a qualitative description of behavior into the controller, it can detect the change in qualitative regime and control the furnace accordingly, providing both faster curing times and higher yield rates than traditional techniques.

Another use of qualitative representations is in describing control strategies used by machine operators, such as unloading cranes on docks. By recording the actions of crane operators, machine learning techniques can be used to reverse-engineer their strategies (Suc and Bratko, 2002).

In some applications a small set of fault models can be pre-enumerated. A set of models, which includes the nominal model of the system plus models representing common faults, can then be used to track the behavior of a system with a qualitative or semiquantitative simulator. Any fault model whose simulation is inconsistent with the observed behavior can thus be ruled out. Relying on a preexisting library of fault models can limit the applicability of automatic monitoring and diagnosis algorithms. One approach to overcoming this limitation is to create algorithms that require only models of normal behavior. Most consistency-based diagnosis algorithms take this approach. For example, in developing onboard diagnostics for Diesel engines, qualitative representations have been found to be useful as a robust level of description for reasoning and for abstracting away from sensor noise (e.g., Sachenbacher et al., 2000).

One limitation with consistency-based diagnosis is that the ways a system can fail are still governed by natural laws, which impose more constraint than logical consistency. This extra constraint can be exploited by using a domain theory to generate explanations that could account for the problem, via abduction. These explanations are useful because they make additional predictions that can be tested and that also can be important for reasoning about safety in operative diagnosis (e.g., if a solvent tank's level is dropping because it is leaking, then where is the solvent going?). However, in many diagnosis tasks, this limitation is not a concern.

### 35.5.2 Design

Engineering design activities are divided into conceptual design, the initial phase, when the overall goals, constraints, and functioning of the artifact are established, and detailed design, when the results of conceptual design are used to synthesize a constructible artifact or system. Most computer-based design tools, such as computer-aided design (CAD) systems and analysis programs, facilitate detailed design. Yet many of the most costly mistakes occur during the conceptual design phase. The ability to reason with partial information makes qualitative reasoning one of the few technologies that provides substantial leverage during the conceptual design phase (Klenk et al., 2012). Qualitative reasoning can also help automate aspects of detailed design.

One example was Mita Corporation's DC-6090 photocopier (Shimomura et al., 1995), a self-maintenance machine, in which redundant functionality is identified at design time so that the system can dynamically reconfigure itself to temporarily overcome certain faults. An envisionment including fault models, created at design time, was used as the basis for constructing the copier's control software. In operation, the copier

keeps track of which qualitative state it is in, so that it produces the best quality copy it can. More recently, fully automatic, real-time composition of control software for "plug and play" high-end printing systems has been fielded by Xerox, using planning techniques over qualitative compositional models to assemble model-based controllers as parts are added to a system in the field (Fromherz et al., 2003).

In some fields experts formulate general design rules and methods expressed in natural language. Qualitative representations can enable these rules and methods can be further formalized, so that they can be automated. In chemical engineering, for instance, several design methods for distillation plants have been formalized using qualitative representations, and designs for binary distillation plants comparable to those in the chemical engineering research literature have been generated automatically.

Automatic analysis and synthesis of kinematic mechanisms have received considerable attention. Complex fixed-axis mechanisms, such as mechanical clocks, can be simulated qualitatively, and a simplified dynamics can be added to produce convincing animations. Initial forays into conceptual design of mechanisms have been made, for example, using configuration space-based techniques to analyze higher-pair mechanisms (Sacks and Joskowicz, 2010). Qualitative representations are also useful in case-based design, because they provide a level of abstraction that simplifies adaptation (e.g., Faltings, 2001).

Qualitative reasoning also is being used to reason about the effects of failures and operating procedures. Such information can be used in FMEA. For example, potential hazards in a chemical plant design can be identified by perturbing a qualitative model of the design with various faults and using qualitative simulation to ascertain the possible indirect consequences of each fault. Commercial FMEA software using qualitative simulation for electrical system design is now being used commercially in automotive design (Price, 2000).

### 35.5.3 Scientific Modeling

Formalizing the art of modeling is one of the goals of qualitative reasoning, which has led to it being applied to scientific modeling in several ways. One approach is to use modeling ideas from qualitative reasoning to provide languages for organizing mathematical models in ways that support software-assisted model formulation and revision based on comparing numerical data to simulation results (e.g., Bridewell et al., 2006). Another approach is to use spatial aggregation to extract meaningful features from spatially distributed data to provide concise summarizations of structure and function (e.g., Ironi and Tentoni, 2011). Finally, in some domains traditional mathematical models are too fine grained compared to the experimental data available; hence, qualitative models can provide a more natural language for expressing theories. Two examples are ecology, for example, Bredeweg et al. (2008), where qualitative models have been used to help articulate underlying assumptions in modeling, and biology, for example, Batt et al. (2012), where hypotheses about bacterial regulatory networks are being explored via qualitative simulation with model checking. Qualitative models are also useful as an intermediate representation for eliciting and formalizing knowledge. For example, the Agroecological Knowledge Toolkit helps agricultural scientists gather and distill local ecosystem knowledge in field studies (e.g., Cerdan et al., 2012). Another example is the Calvin system (Rassbach et al., 2011), which uses qualitative representations of evidence to help geoscientists construct arguments concerning the interpretation of cosmogenic isotope data in rocks.

### 35.5.4 Intelligent Tutoring Systems and Learning Environments

One of the original motivations for the development of qualitative reasoning was its potential applications in intelligent tutoring systems (ITSs) and intelligent learning environments (ILEs). Qualitative representations provide a formal language for a student's mental models (Gentner and Stevens, 1983), and thus they facilitate communication between software and student. For example, if a student modeling a block on a table leaves out the force the table exerts on the block, they can be shown a simulation where the block falls through the table, causing them to improve their model (Hirashima et al., 2009). Similarly, in learning organic chemistry, qualitative representations can provide explanations for what happens in reactions (Tang et al., 2010). For younger students, who have not had algebra or differential equations,

the science curriculum consists of learning causal mental models that are well captured by the formalisms of qualitative modeling. By using a student-friendly method of expressing models, such as concept maps, software systems have been built that help students learn conceptual models (Forbus et al., 2005; Leelawong et al., 2001), providing software support that scaffolds students in modeling (Bredeweg et al., 2010). In more advanced courses and in industrial training, qualitative reasoning provides explanations that help students make sense of the results of simulators and quantitative analysis tools, for example, to explain why a student's design violates the laws of thermodynamics when teaching about power plants.

### 35.5.5 Cognitive Modeling

Since qualitative reasoning was inspired by observations of how people reason about the physical world, one natural application of qualitative reasoning is cognitive simulation, that is, the construction of programs whose primary concern is accurately modeling some aspect of human reasoning and learning, as measured by comparison with psychological results. Some research has been concerned with modeling scientific discovery and conceptual change, for example, simulating the trajectories of mental models that students go through when learning intuitive notions of force (Friedman and Forbus, 2010). Several investigations suggest that qualitative representations have major role to play in understanding cognitive processes such as high-level vision. For example, Sridhar et al. (2010) describes how qualitative spatial representations can be used to help analyze video data, and Lovett et al. (2010) describes how analogical reasoning over automatically computed qualitative representations suffices to enable a computational model to outperform most adult Americans on Raven's Progressive Matrices, a visual reasoning test that correlates well with general intelligence. Qualitative order of magnitude representations play a central role in a model of moral decision-making (Dehghani et al., 2008), providing the stratification needed to model the effects of sacred values in decision-making. Commonsense reasoning appears to rely heavily on qualitative representations, although human reasoning may rely more on reasoning from experience than first-principle reasoning (Forbus and Gentner, 1997). Understanding the robustness and flexibility of human commonsense reasoning is an important scientific goal in its own right and will provide clues as to how to build better AI systems. Thus, potential use of qualitative representations by cognitive scientists may ultimately prove to be the most important application of all.

## 35.6 Research Issues and Summary

Qualitative reasoning is now a mature subfield with a mixture of basic and applied activities, including a number of fielded applications. Its importance is likely to continue to grow, due to the increasing number of successful applications, the continued growth in available computing power, and the rapid digitization of workflows in many human endeavors.

Although there is a substantial research base to draw upon, there are many open problems and areas that require additional research. There are still many unanswered questions about purely qualitative representations (e.g., what is the minimum information that is required to guarantee that all predicted behaviors generated from an envisionment are physically possible?), but the richest vein of research concerns the integration of qualitative knowledge with other kinds of knowledge: numerical, analytic, teleological, etc. The work on modeling to date, although a solid foundation, is still very primitive; better model formulation algorithms, well-tested conventions for structuring domain theories, and robust methods for integrating the results of multiple models are needed. Most domain modeling efforts have been driven by particular application domains and tasks, which is effective in the short run, but a more open development strategy would speed the development of integrated domain theories for a broad range of scientific and engineering knowledge. The growing importance of deep natural language understanding (Ferrucci et al., 2012) is likely to invigorate research in exploring the role of qualitative representations in natural language semantics, since they provide a natural level of description for everyday properties and events (Kuehne and Forbus, 2004; Pustejovsky and Moszkowicz, 2011).

## Key Terms

**Comparative analysis:** A particular form of a "what-if" question, that is, how a physical system changes in response to the perturbation of one of its parameters.

**Compositional modeling:** A methodology for organizing domain theories so that models for specific systems and tasks can be automatically formulated and reasoned about.

**Confluence:** An equation involving sign values.

**Diagrammatic reasoning:** Spatial reasoning, with particular emphasis on how people use diagrams.

**Domain theory:** A collection of general knowledge about some area of human knowledge, including the kinds of entities involved and the types of relationships that can hold between them, and the mechanisms that cause changes (e.g., physical processes and component laws). Domain theories range from purely qualitative to purely quantitative to mixtures of both.

**Envisionment:** A description of all possible qualitative states and transitions between them for a system. Attainable envisionments describe all states reachable from a particular initial state; total envisionments describe all possible states.

**FMEA (Failure Modes and Effects Analysis):** The analysis of the consequences of the failure of one or more components for the operation of an entire system.

**Landmark:** A comparison point indicating a specific value achieved during a behavior, for example, the successive heights reached by a partially elastic bouncing ball.

**Limit point:** A comparison point indicating a fundamental physical boundary, such as the boiling point of a fluid. Limit points need not be constant over time, example, boiling points depend on pressure.

**Metric diagram:** A quantitative representation of shape and space used for spatial reasoning, the computer analog to, or model of, the combination of diagram/visual apparatus used in human spatial reasoning.

**Model fragment:** A piece of general domain knowledge that is combined with others to create models of specific systems for particular tasks.

**Modeling assumption:** A proposition expressing control knowledge about modeling, such as when a model fragment is relevant.

**Physical process:** A mechanism that can cause changes in the physical world, such as heat flow, motion, and boiling.

**Place vocabulary:** A qualitative description of space or shape that is grounded in a quantitative representation.

**Qualitative proportionality:** A qualitative relationship expressing partial information about a functional dependency between two parameters.

**Qualitative simulation:** The generation of predicted behaviors for a system based on qualitative information. Qualitative simulations typically include branching behaviors due to the low resolution of the information involved.

**Quantity space:** A set of ordinal relationships that describes the value of a continuous parameter.

**Semiquantitative simulation:** A qualitative simulation that uses quantitative information, such as numerical values or analytic bounds, to constrain its results.

## Further Information

Several publically available qualitative reasoning systems that can be downloaded from the web provide an easy way to get started. Those currently being actively maintained and extended include University of Amsterdam's Garp3, a qualitative dynamics simulator; University of Bremen's SparQ; a qualitative spatial reasoning engine; and Northwestern University's CogSketch, a sketch understanding system.

Papers on qualitative reasoning routinely appear in AI, Journal of Artificial Intelligence Research (JAIR), and IEEE Intelligent Systems. Many papers first appear in the proceedings of the American